PRECALCULUS · UNIT 7: VECTORS

Flight Path Navigator

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students plan flights for an island air delivery service. Map positions become vectors, wind and airspeed add to give ground velocity, and dot products find angles, projections, and work.

Wind correction shows why vector addition matters in real navigation.

Standards Addressed

- N-VM.A.1–3 Vector quantities, components, magnitude, direction
- N-VM.B.4–5 Add, subtract, and scale vectors
- Dot product, angle between vectors, and projections
- Applications: navigation and work

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a flight-tracker screenshot. Part 1 map vectors.
Day 2	Part 2 wind and headings.
Day 3	Part 3 dot product, work, and the flight plan.

Materials

- Student handout and Flight Data Cards
- Graph paper and protractors
- Scientific calculators
- Optional: online vector applet

How the Three Versions Differ

Level	What changes
CP2	Component form, magnitude, direction, a simple wind resultant, and scalar multiples.
CP1	Route closure, bearings, wind correction with a heading, dot-product angles, and work.
Honors	Diagonal winds, heading verification, tailwind and crosswind projections, ramp forces, and route comparison.

Watch For (Common Misconceptions)

- Mixing bearings (from north, clockwise) with standard angles (from east, counterclockwise).
- Adding magnitudes instead of vectors.
- Using sine instead of cosine for work.

Extension Ideas

- Use real wind forecasts for a flight.
- Explore boats crossing a river with current.
- Program a vector calculator in a spreadsheet.

Answer Key – CP2

Part 1: Vectors on the Map

#	Task	Answer
1	Write the vector from the Mainland Airport to Gull Island in component form.	$\langle 120, 90 \rangle$
2	Find its magnitude (the distance in miles).	$\sqrt{(120^2 + 90^2)} = 150$ miles
3	Find its direction angle from east.	$\tan^{-1}(\frac{90}{120}) \approx 36.9°$ north of east
4	Write the vector from Gull Island to Pine Island.	$\langle 80, -140 \rangle$

Part 2: Wind

#	Task	Answer
5	Write the plane's air velocity as a vector.	$\langle 0, 150 \rangle$
6	Add the wind vector to find the ground velocity.	$\langle 30, 150 \rangle$
7	Find the ground speed.	$\sqrt{(900 + 22{,}500)} \approx 153.0$ mph
8	By how many degrees does the wind push the plane off course?	$\tan^{-1}(\frac{30}{150}) \approx 11.3°$ east of north

Part 3: Plan a Flight

#	Task	Answer
9	If the ground velocity is $\langle 30, 150 \rangle$ mph, where is the plane after 2 hours?	$2\langle 30, 150 \rangle = \langle 60, 300 \rangle$
10	Find a unit vector in the direction of Gull Island.	$\langle \frac{120}{150}, \frac{90}{150} \rangle = \langle 0.8, 0.6 \rangle$
11	Draw a vector diagram of one flight and label the air velocity, wind, and ground velocity.	Answers vary.

Reflection

#	Prompt	Look for
R1	What does the resultant vector represent for the plane?	Its actual motion over the ground (speed and direction).
R2	Why can the ground speed be faster than the airspeed?	A tailwind or crosswind component adds to the plane's velocity.

Answer Key – CP1

Part 1: Route Vectors

#	Task	Answer
1	Find the vectors and distances for Mainland → Gull → Pine → Mainland.	⟨120, 90⟩ 150 mi; ⟨80, −140⟩ ≈ 161.2 mi; ⟨−200, 50⟩ ≈ 206.2 mi
2	Show that the three vectors add to the zero vector. Why?	⟨0, 0⟩; the route returns to the start.
3	Find the bearing (clockwise from north) from Mainland to Gull Island.	90° − 36.9° ≈ 053°
4	Find the angle at Gull Island between the incoming and outgoing legs using the dot product.	Vectors G→M ⟨−120, −90⟩ and G→P ⟨80, −140⟩: $\cos\theta = \frac{-9{,}600+12{,}600}{150\times 161.2} \approx 0.124 \rightarrow \theta \approx 82.9°$

Part 2: Wind Correction

#	Task	Answer
5	If the pilot aims straight at Gull Island, what is the ground velocity and where does the plane actually go?	Air ⟨120, 90⟩ (150 mph toward Gull) + ⟨30, 0⟩ = ⟨150, 90⟩; it drifts east of Gull Island
6	Find the heading so the ground track points at Gull Island. (Let the air velocity be 150⟨cos θ, sin θ⟩.)	The ground velocity ⟨150 cos θ + 30, 150 sin θ⟩ must be parallel to ⟨120, 90⟩: 0.6(150 cos θ + 30) = 0.8(150 sin θ). Solving (graph or table) gives θ ≈ 43.8° from east, a heading of about 046°.
7	Find the ground speed with that heading and the flight time.	Ground ≈ ⟨138.3, 103.7⟩ → ≈ 173 mph; 150 ÷ 173 ≈ 0.87 h ≈ 52 min
8	Explain why the corrected heading points into the wind.	The plane must aim partly against the wind so the wind's push brings it back on track.

Part 3: Cargo & Work

#	Task	Answer
9	A worker pulls a cart with 80 lb of force at 25° above the ramp for 40 ft. Find the work.	W = 80 × 40 × cos 25° ≈ 2,900 ft·lb
10	Are ⟨3, 4⟩ and ⟨−8, 6⟩ perpendicular? Use the dot product.	−24 + 24 = 0; yes
11	Write your flight plan for the full route with headings, ground speeds, and times.	Answers vary.

Reflection

#	Prompt	Look for
R1	How does the dot product reveal the angle between two vectors?	$u \cdot v = \|u\|\|v\| \cos\theta$, so $\cos\theta = \{u \cdot v\|\|u\|\|v\|\}$.
R2	What is the difference between heading and course?	Heading is where the plane points; course is where it actually travels over the ground.

Answer Key – Honors

Part 1: Wind Correction

#	Task	Answer
1	Write Wind B in component form.	$25\langle \sin 135°, \cos 135° \rangle \approx \langle 17.68, -17.68 \rangle$
2	Find the heading needed to fly from Gull Island to Pine Island (direction $\langle 80, -140 \rangle$) in Wind B.	Course bearing of $\langle 80, -140 \rangle \approx 150.3°$. The crosswind part of Wind B is about 6.6 mph, pushing the plane to the left (east) of the course, so turn into it by $\sin^{-1}\left(\frac{6.6}{150}\right) \approx 2.5°$: heading ≈ 152.8°.
3	Find the ground speed and time for the leg.	Along-track speed ≈ 150 cos 2.5° + 24.1 ≈ 174.0 mph; 161.2 ÷ 174.0 ≈ 0.93 h ≈ 56 min
4	Verify your heading by adding vectors.	Air $150\langle \sin 152.8°, \cos 152.8° \rangle \approx \langle 68.6, -133.4 \rangle$ plus wind $\langle 17.7, -17.7 \rangle = \langle 86.3, -151.1 \rangle$; $\frac{-151.1}{86.3} \approx -1.75 = \frac{-140}{80}$, so it is parallel to the course ✓

Part 2: Projections

#	Task	Answer
5	Find the component of Wind A $\langle 30, 0 \rangle$ along the route to Gull Island $\langle 120, 90 \rangle$. Is it a tailwind?	$\frac{3{,}600}{22{,}500}\langle 120, 90 \rangle = \langle 19.2, 14.4 \rangle$ (magnitude 24 mph); yes, a tailwind
6	Find the crosswind component (perpendicular part).	$\langle 30, 0 \rangle - \langle 19.2, 14.4 \rangle = \langle 10.8, -14.4 \rangle$ (18 mph)
7	A 600-lb crate sits on a 15° ramp. Find the component of its weight that pulls it down the ramp.	600 sin 15° ≈ 155.3 lb
8	How much force parallel to the ramp is needed to pull the crate 40 ft up the ramp, and how much work is done?	≈ 155.3 lb; W ≈ 6,212 ft·lb

Part 3: Route Optimization

#	Task	Answer
9	Compare the total distance of Mainland → Gull → Pine → Mainland with Mainland → Pine → Gull → Mainland.	Both ≈ 517.4 mi (same triangle, reversed)
10	With Wind A $\langle 30, 0 \rangle$, which direction around the triangle takes less total time? For each leg, use ground speed ≈ $\sqrt{(150^2 - \text{crosswind}^2)}$ + tailwind.	With wind-corrected headings: M→G ≈ 52 min, G→P ≈ 59.5 min, P→M ≈ 102.5 min (total ≈ 214 min); reverse: M→P ≈ 69.1, P→G ≈ 72.8, G→M ≈ 72.0 (total ≈ 214 min). The totals are equal; the direction only changes which legs are fast or slow.

#	Task	Answer
11	Write the final flight plan with headings, ground speeds, times, and a vector diagram.	Answers vary.

Reflection

#	Prompt	Look for
R1	Why does splitting the wind into along-track and crosswind parts simplify navigation?	The crosswind must be cancelled by the heading; the along-track part changes the ground speed.
R2	How is the work done related to projection?	Work uses only the force component along the displacement, which is the projection.